Predicting Hispanic Student Six-Year Completion Rates at U.S. Four-Year Institutions Using IPEDS Data

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I. INTRODUCTION

Institutional success is no longer determined by the number of prospective students who enroll, but also by retention, and ultimately, their completion rates. Completing higher education is an important outcome, as it leads to better employment opportunities and higher paying jobs. As a result, analyzing the factors that may influence completion rates has become increasingly relevant, particularly when examining how outcomes vary across different student demographics. While predicting completion rates may pose complexity, since the outcomes are influenced by multiple factors that interact in various ways.

Within the growing population of degree-seeking individuals, Hispanic students represent one of the fastest growing groups pursuing higher education within the U.S. Between fall 2010 and fall 2021, enrollment for Hispanic students increased by 30%, showing a growth from 2.6 million to 3.3 million [1]. Within U.S. demographics, Hispanics are the second largest group of undergraduate student enrollment. In fall 2021 of the 15.4 million students enrolling in undergraduate programs, 3.3 million were Hispanic [1]. Despite this increase in enrollment, completion rates for Hispanic students are below the national average, and reflect a significant completion rate gap relative to White and Asian peers. For instance, based on the 2010 cohort of students, the national six-year completion rate trend for Hispanic students reflected 54%, while Asian students reflected 74%, and White students 64% [2].

These differences may be influenced by several institutional factors, such as public and/or private institutions, financial resources and financial aid, retention rates, admissions criteria, and other resources available to students. Hispanic students represent an important and growing demographic group in higher education. It would be beneficial for institutions to better understand the factors that influence their ability to successfully complete a degree. Yet, it is not always clear which factors are most significant in these completion rate outcomes. Therefore, identifying the factors that strongly influence Hispanic student completion rates is an important problem, and determining them in order to predict future completion rates.

This project uses institutional data from the National Center for Education Statistics (NCES) Integrated Postsecondary Education Data System (IPEDS) to examine and model Hispanic student bachelor's degree completion rates within a six-year time frame (150% of the four-year normal time). The dataset provides information on institutional characteristics, academic, and financial variables from U.S colleges and universities. Using this data, the goal of the project is to develop a statistical model that can predict college completion rates for Hispanic students based on these variables.

II. STATE-OF-THE-ART

A. Measuring "completion" rates: why the definition matters?

When predicting Hispanic bachelor's degree completion using institution-level data, the majority of the studies in the U.S. are based on the same backbone: IPEDS. IPEDS graduation-rate measures are widely used because they are standardized across institutions and published on federal dashboards, but they also come with well-known blind spots. The '150% time' metric (six years for a bachelor's) focuses on first-time, full-time entrants, and treats students who transfer out as noncomplete at their starting institution, even if they later finish elsewhere. That is a practical choice for accountability reporting, but it complicates interpretation when we are trying to describe equity gaps or institutional effectiveness [2].

Given that Hispanic students are disproportionately represented among students who start at broad-access institutions, attend part-time, or move between colleges, these decisions on measurements can understate completion pathways that involve transferring. This is one reason the state of the art has gradually shifted from treating raw graduation rates as 'the answer' to treating them as one indicator that needs context, especially when comparing institutions with very different student mixes [3].

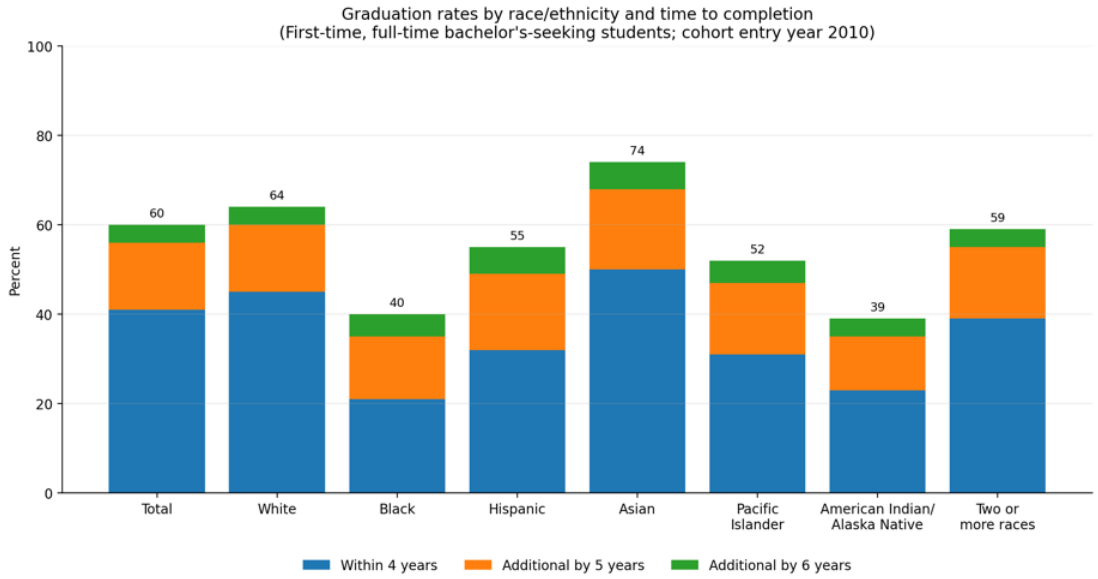

Fig. 1. Graduation rates by race/ethnicity and time to completion (cohort entry year 2010), based on data from NCES Indicator 23, *Postsecondary Graduation Rates* [2].

Even with those caveats, the national patterns are consistent across data sources: Hispanic students complete bachelor's degrees at lower rates than White and Asian students on average. In the cohort shown in Fig. 1, the six-year graduation rate from the first institution attended was 54% for Hispanic students versus 64% for White students and 74% for Asian students [2]. A common takeaway in recent literature is that 'gap' language is too simplistic because it blends together student-level barriers (finances, time, family obligations) and institutional factors (resources, selectivity, advising capacity). That split matters if the goal is not just describing disparities, but understanding what barriers institutions actually control.

B. Institutional context: creating "adjusted" comparisons.

Another thread in the literature focuses on the institutional sector. Public and private nonprofit institutions graduate higher shares of first-time, full-time students than the for-profit sector, and the pattern holds across different racial/ethnic groups [2]. That basic fact shows up quickly in descriptive work, but it also shapes how researchers build models: it is difficult to interpret an institution's graduation rate without accounting for selectivity, student composition, and spending.

One influential recent example is the 'Bending the Curve' approach, which treats graduation as something to be predicted from institutional characteristics and student-body composition, and then uses the residual (actual minus predicted) as a more apples-to-apples performance indicator. Using a decade of IPEDS data, de Castro Galvao et al. [3] show that a large share of between-college variation is explained by compositional factors and instructional spending; meaning that raw graduation-rate rankings can be misleading if taken as a direct measure of institutional effectiveness.

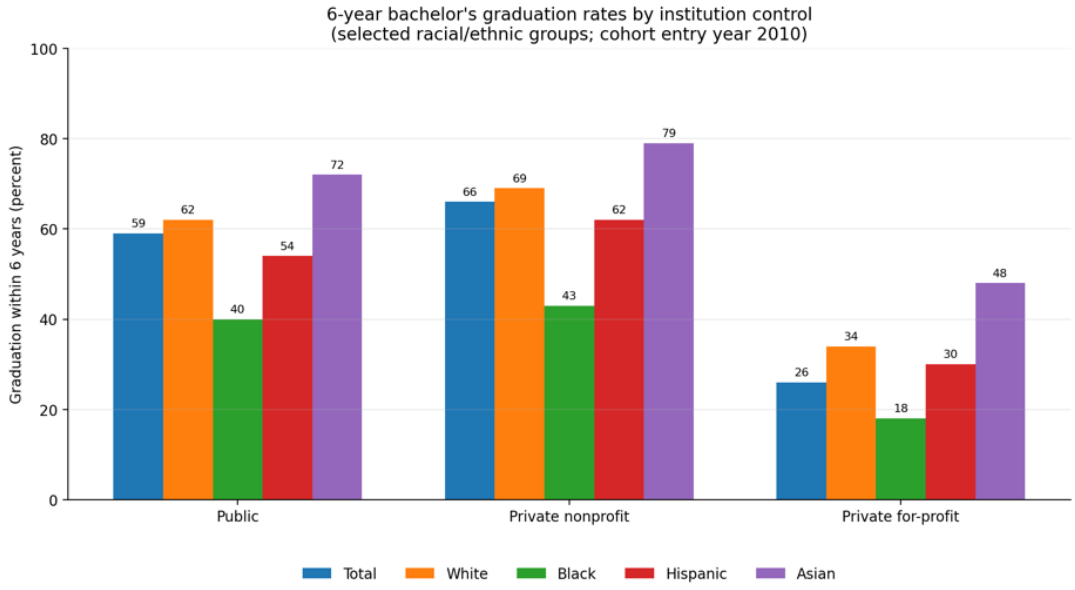

Fig. 2. Six-year bachelor's graduation rates by institution control (selected groups; cohort entry year 2010), based on NCES Indicator 23 (Fig. 23.3) [2].

The modeling implication displays a clear method: if the goal is prediction, then sector and resource variables usually enter early and often; if the goal is accountability, then researchers increasingly argue for adjusted benchmarks rather than raw rates. In practice, many institutional-effectiveness studies with IPEDS data still rely on linear or generalized linear models because they are transparent and interpretable, but they augment those models with richer controls and sensitivity checks [3].

C. Predictive modeling: what can be learned from the ML comparisons?

Over the last few years, the 'prediction' side of completion research has grown fast. Institutions increasingly use early-alert and risk-scoring systems, and the research community has responded by comparing conventional regression models against more complex machine-learning approaches. Bird et al. [4] large comparison study is often cited here because it finds that careful sample and feature construction

can matter as much as the choice between simple and complex models, and that, in many settings, basic logistic/linear models perform similarly to more sophisticated methods while remaining more interpretable.

At the same time, the state of the art has become more skeptical about 'better prediction' as an unqualified good. A newer body of work focuses on algorithmic bias: models trained on historical data can systematically underestimate success for racially minoritized students, even when race is not used explicitly. For example, Gándara et al. [5] show that student-success prediction errors are not evenly distributed across racial/ethnic groups, and common bias-mitigation techniques reduce but do not fully eliminate disparities.

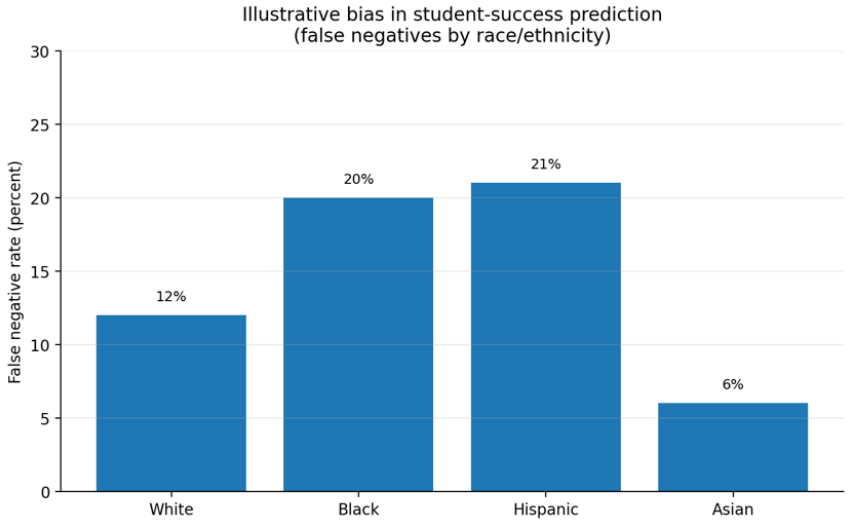

Fig. 3. Illustrative example of unequal error rates in student-success prediction models, based on false-negative rates reported in [5] and summarized in [6].

Even though our project focuses on institution-level outcomes rather than individual student risk scores, the lesson transfers. If a model will be used for comparison or resource allocation, we need to pay attention to who the model systematically over- or under-predicts. That is why many recent papers recommend pairing accuracy metrics with subgroup diagnostics and clear documentation of modeling choices [4], [5].

Putting these strands together, the current 'best practice' looks less like a single method and more like a checklist: use standardized federal measures (usually IPEDS) but acknowledge what they miss; model graduation in a way that separates compositional factors from institutional capacity; and, if predictive models are used for decisions, evaluate them for transparency and equity. This project sits right at that intersection, using institution-level predictors to understand variation in Hispanic completion while still treating completion rates as a measure that needs context.

III. MATERIALS AND METHODS

A. Dataset

All data were obtained from NCES IPEDS, academic year 2023–24 release [https://nces.ed.gov/ipeds]. Six IPEDS files were used to assemble the modeling dataset: GR2024

(Graduation Rates) used to construct the target variable, IC2024 (Institutional Characteristics) providing institution type, EF2024D (Fall Enrollment) providing retention rate and student-faculty ratio, SFA2324 (Student Financial Aid) providing Pell percentage, loan percentage, and average loan amount, HD2024 (Institutional Directory) providing institution name and state, and ADM2024 (Admissions) providing admissions amounts and SAT score percentiles.

Target variable. IPEDS Graduation Rates data are reported in long format, where each row represents a specific statistic identified by a line code. For Hispanic students in the 2024 dataset, two lines codes are required to construct the completion metric: LINE "10" represents the Hispanic student cohort size, and LINE "29A" represents the number of Hispanic students completing within 150% of normal program time. For four-year institutions, completion within 150% of normal time corresponds to six-years. The Hispanic six-year bachelor's degree completion rate (latino_completion_rate), also to be referred as Hispanic completion rate, was calculated as the ratio of Hispanic six-year completions (LINE 29A) to the Hispanic cohort size (LINE 10), multiplied by 100 to get the result as a percentage.

To ensure reliable estimates, three filtering rules were applied: (1) the dataset was restricted to four-year institutions using institutional level variable (LEVEL1 = 1); (2) institutions with Hispanic cohort size below 15 students were excluded to avoid unstable completion rates; and (3) observations with completion rates outside the valid range [0, 100] were excluded. The final modeling dataset contains approximately 1,505 institutions.

Predictors. Predictor variables were also constructed, these variables were mainly academic, financial or institutional. Academic predictors such as retention rates (retention_rate), student-faculty ratio (student_faculty_ratio), admission rate (admission_rate), and SAT median (sat_median). Financial predictors such as percent of students receiving Pell Grants (pct_pell), percent of students receiving loans (pct_loan), and average loan amount (avg_loan_amt). Institutional predictor was institution type (CNTLAFFI), a categorical variable categorized by the IPEDS as (1) Public, (2) Private for-profit, (3) Private not-for-profit (no religious affiliation), and (4) Private not-for-profit (religious affiliation).

TABLE I. FEATURES USED IN THE MODELING DATASET

Feature	Description
latino_completion_rate	Six-year completion rate for Hispanic students (%)
retention_rate	First-year student retention rate (%)
student_faculty_ratio	Number of students per faculty member
pct_pell	% of students receiving pell Grants
pct_loan	% of students receiving student loans
avg_loan_amt	Average student loan amount (USD)
admission_rate	Admission rate (ratio of admitted to applicants)
sat_median	Median SAT score (combined math and verbal)
sat_missing	Missing SAT indicator (1 = missing)
admission_missing	Indicator for missing admission data (1 = missing)
CNTLAFFI	Institutional control type (public/private categories)

Data cleaning and preprocessing. Several preprocessing steps were performed to ensure consistency. First, IPEDS frequently uses coded values such as (-1, -2, and -3) to represent missing or suppressed data. Therefore, the coded values were converted to standard missing values (NA). Second, train only median imputation was applied to continuous variables including retention rate, student-faculty ratio, percent Pell Grants, percent loans, average loan amount, admission rate, and SAT median. This approach allows institutions with small amounts of missing information to remain in the analysis while minimizing bias introduced by missing values. Third, binary indicators were created for each of the previously mentioned continuous variables. These indicators capture if the original value was missing before, allowing the model to differentiate between institutions with typical selectivity levels and institutions that do not report selectivity metrics. Lastly, treating institutional type as a numeric variable would incorrectly impose a linear ordering on categorical institution types. To avoid this, the variable was encoded as a factor, allowing the model to estimate separate effects for each category relative to a reference group.

B. Methods

Five modeling techniques were applied to the cleaned dataset.

1) Multiple Linear Regression: Model was fit on the training subset using all predictors and used as an interpretable baseline.

2) Elastic Net Regression: Model was fit at two parameters: $\alpha = 0.5$ (balanced ridge-lasso combination) and $\alpha = 0.1$ (ridge-leaning). The regularization strength λ was selected for each via 10-fold cross-validation.

3) Random Forest: Model was fit with 500 trees and default mtry. Variable importance was computed via two measures: percentage increase in mean squared error under permutation (%IncMSE) and total reduction in node purity (IncNodePurity).

4) Bagging: Model was implemented as a random forest with mtry = p (i.e., all predictors considered at every split, eliminating the random feature subsampling component of standard random forests), 500 trees.

5) Gradient Boosting Machine (GBM): Model was fit with Gaussian loss, 5,000 trees, interaction depth 2, and shrinkage 0.01. The optimal number of boosting iterations was selected by 10-fold cross-validation, the procedure selected optimal iteration of 1,036, and that value was used for all test-set predictions.

C. Evaluation

All models were trained and evaluated on a single 70/30 hold-out split (set.seed = 1) of the 1,505-institution dataset, producing 1,053 training and 452 test observations. Hyperparameters λ for elastic net and the number of boosting iterations for gradient boosting machine (GBM), were selected via 10-fold cross-validation within the training set only. Test-set performance was reported using three metrics: mean squared error (MSE, average squared deviation between predicted and actual completion rates), root mean squared error (RMSE, in percentage-point units of the completion rate), and the coefficient of determination (R^2, proportion of test-set variance explained). Pairwise differences between models were tested using paired t-tests on per-observation squared errors, the paired structure controls for the fact that all five models predict the same test observations.

IV. RESULTS

A. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the distribution of Hispanic completion rates and to identify patterns and relationships between the key predictors and target variables prior to modeling. The Hispanic six-year completion rate ranged from 5.6% to 100% across the dataset, with a median of 41.4% and a mean of 44.8%. The distribution is shown in Fig. 1.

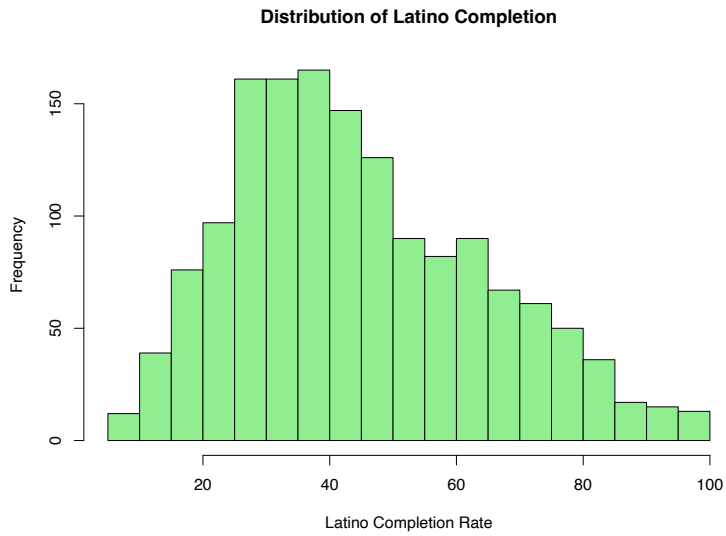

Fig. 4. Distribution of Hispanic six-year completion rates across U.S. four-year institutions in the IPEDS 2024.

A correlation analysis of each predictor variable against the target identified retention rate as the strongest individual predictor (+0.46), followed by average loan amount (+0.41) and percent loans (+0.40). SAT median score (+0.32) and admission rate (−0.27) showed moderate correlations. Notably, percent Pell Grants exhibited essentially no linear correlation with the target (−0.04), yet, later analyses will reflect it as one of the stronger predictors in the random forest model, signaling the importance of nonlinear methods for this data. Sorted correlations are shown in Table II.

TABLE II. PEARSON CORRELATIONS WITH THE HISPANIC COMPLEATION RATE

Feature	Pearson Correlation (r)
retention_rate	+0.461
avg_loan_amt	+0.407
pct_loan	+0.401
CNTLAFFI (numeric)	+0.393
sat_median	+0.324
pct_pell	−0.040
student_faculty_ratio	−0.092
sat_missing	−0.248

admission_rate	−0.274
admission_missing	−0.286

The relationship between retention rate and completion rate is strongly positive. Scatter plot of retention rate and completion rate shown in Fig. 5. The plot suggests a median retention rate near 70%. However, a concentration of observations at a specific retention value is observed due to median imputation of missing data, which introduces artificial clustering without affecting the overall trend. Additionally, institutions above this median retention level tend to show substantially higher Hispanic completion rates, reinforcing retention as one of the strongest predictors of graduation outcomes.

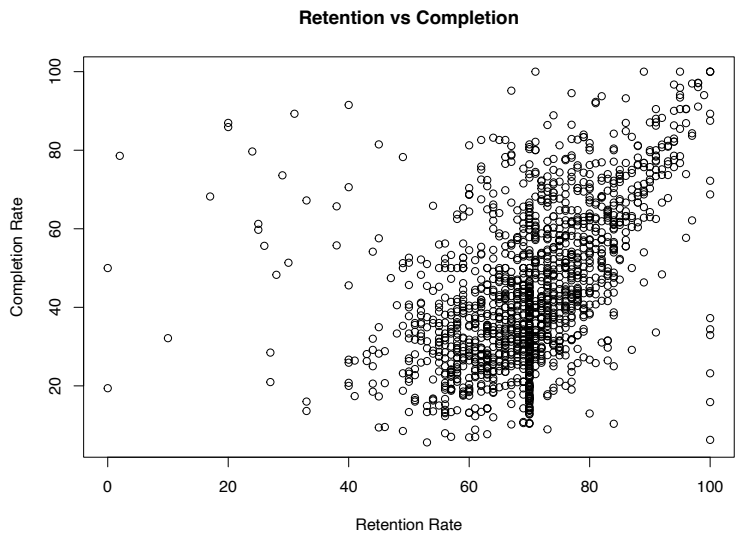

Fig. 5. First-year retention rate vs. six-year Hispanic completion rate (r = +0.46), from the IPEDS 2024

B. Linear Regression

Multiple linear regression was fit on the 1,053 training observations using all predictors. On the test set, the model achieved MSE = 186.49, RMSE = 13.66, and $R^2 = 0.483$. Diagnostic plots Fig. 6 show a slight right in the residuals and a few outliers, but overall assumes linearity.

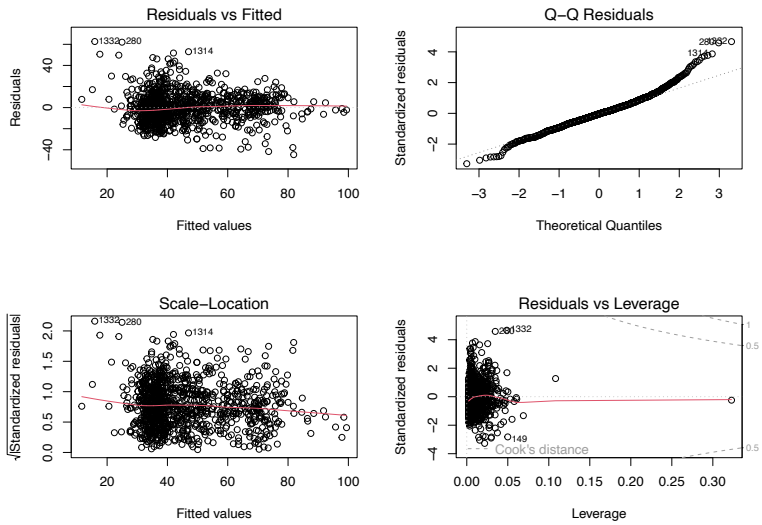

Fig. 6. Diagnostic plots for the linear regression model: residuals vs fitted values, Q-Q plot of standardized residuals, scale-location plot, and residuals vs leverage.

Coefficient estimates are summarized in Table III. The strongest positive predictors were retention rate and SAT median score. For institution type, three of the four levels, signal strong positive significance relative to public institutions. Notably, the coefficient on private for-profits (CNTLAFFI2) was much larger than either of the not-for-profit coefficients (CNTLAFFI3 and CNTLAFFI4), indicating that institution type effects are dominated by the for-profit category within private institutions. Admission rate had a significant negative coefficient, which aligns with the interpretation that more selective institutions result in higher Hispanic completion rates. While, percent Pell Grants and percent loans were not statistically significant within the linear model.

TABLE III. LINEAR REGRESSION COEFFICIENT ESTIMATES

Feature	Estimate	Std. Error	t-value	p-value	Sig.
(Intercept)	-20.67578	1.20E+01	-1.7275735	8.44E-02	.
retention_rate	0.34517322	4.18E-02	8.2536296	4.62E-16	***
student_faculty_ratio	-0.2168215	6.19E-02	-3.5010316	4.83E-04	***
pct_pell	-0.0111624	3.27E-02	-0.3410413	7.33E-01	
pct_loan	-0.034154	2.40E-02	-1.4240254	1.55E-01	
avg_loan_amt	0.00055452	2.43E-04	2.2786039	2.29E-02	*
admission_rate	-14.795799	4.62E+00	-3.2030379	1.40E-03	**
sat_median	0.09881561	1.66E-02	5.9676077	3.30E-09	***
sat_missing	-3.9020099	1.58E+00	-2.4626403	1.40E-02	*
admission_missing	-4.8922023	1.68E+00	-2.92036	3.57E-03	**
CNTLAFFI2	31.943962	1.98E+00	16.103115	2.94E-52	***
CNTLAFFI3	5.31498345	1.98E+00	2.679328	7.49E-03	**
CNTLAFFI4	5.5659667	1.73E+00	3.2100514	1.37E-03	**

Note: Estimates from the linear regression fit on the training set (n = 1,053). Reference category for CNTLAFFI is 1 (Public). Significance codes: *** $p < 0.001$; ** $p < 0.01$; * $p < 0.05$; . $p < 0.1$.

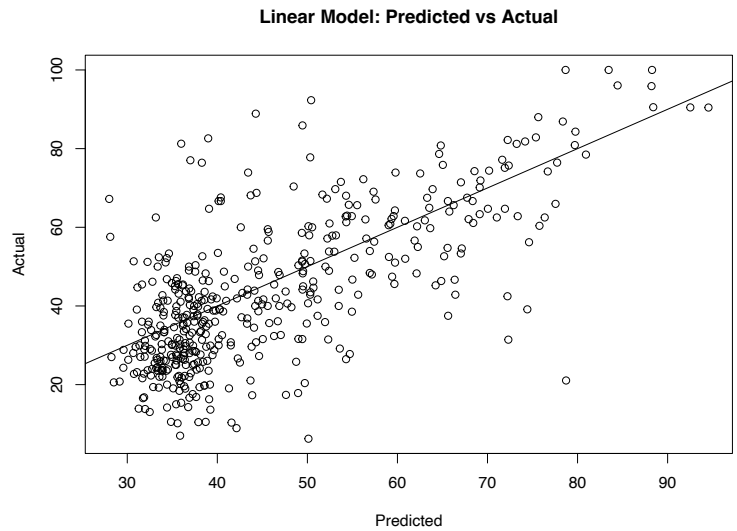

Fig. 7. Test-set predictions vs actual completion rates for the linear regression model.

The predicted versus actual completion rates plot provides a visual assessment of the linear model's performance. Fig. 6 demonstrates a clear positive relationship, suggesting the model is effectively capturing the overall trend in the data. Many observations are clustered around the diagonal line, indicating reasonably accurate predictions for a large portion of institutions. However, there is noticeable dispersion around the line, particularly in the mid-range of predicted values. This spread indicates that while the model captures general patterns, there is still variability that is not fully explained. Additionally, some outliers are present, where the model either overestimates or underestimates completion rates.

C. Elastic Net

To assess whether the linear regression model suffered from overfitting or multicollinearity, a regularized linear model elastic net was evaluated. Two elastic net parameters were compared: $\alpha = 0.5$ (a balanced ridge–lasso mix) and $\alpha = 0.1$ (ridge-leaning). Testing a ridge leaning parameter is desirable in this context because Ridge regression shrinks coefficients without eliminating variables entirely, preserving potentially important predictors, $\alpha = 0.1$ emphasizes coefficient shrinkage while minimizing feature removal. Both achieved nearly identical test performance: at $\alpha = 0.5$, MSE = 186.39, RMSE = 13.65, $R^2 = 0.483$; at $\alpha = 0.1$, MSE = 186.15, RMSE = 13.64, $R^2 = 0.484$. The $\alpha = 0.1$ parameter was retained for later comparison as the result was slightly better.

Model performance was evaluated using cross-validation, a resampling technique that repeatedly splits the data into training and validation sets to provide a reliable estimate of out-of-sample error. This approach ensures that model evaluation is not dependent on a single train-test split and helps identify the optimal level of regularization. The cross-validation curve for $\alpha = 0.1$ is shown in Fig. 7, the curve flattens across a broad range of λ values, indicating that the choice of regularization strength has minimal impact on test performance for this data.

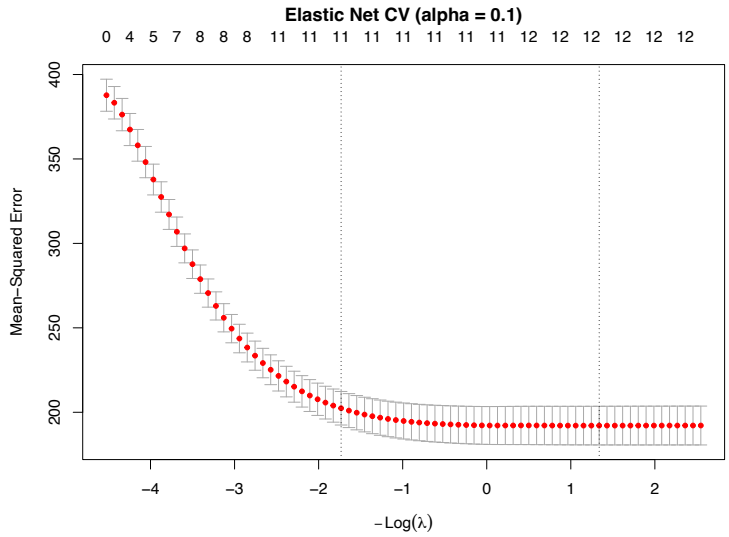

Fig. 8. Cross-validation curve for the Elastic Net model with α = 0.1

Elastic net model resulted in nearly identical results to the linear regression model. This indicates that the linear regression model was not significantly overfitting, and that regularization may have removed useful signal from the predictors rather than improving generalization. Given the relatively small number of features and the stability of the coefficient estimates, regularization did not provide a clear benefit in this case.

D. Random Forest

A machine learning technique was incorporated to evaluate whether a more flexible model could improve predictive accuracy. The random forest model was implemented to capture potential nonlinear relationships and interaction effects that cannot be modeled by linear regression. It is an ensemble method that constructs multiple decision trees and aggregates their predictions, allowing it to model complex patterns in the data.

The random forest model with 500 trees and default *mtry* was fit on the training set and produced the best test-set performance of any model considered: MSE = 159.18, RMSE = 12.62, R^2 = 0.559. This represents a 7.6% reduction in RMSE and a 7.6 percentage-point gain in R^2 over linear regression model.

The variable importance plot from the random forest model provides insight into which predictors contribute most to the model's predictive performance. Two measures are presented: the percentage increase in mean squared error (%IncMSE) and the increase in node purity (IncNodePurity).

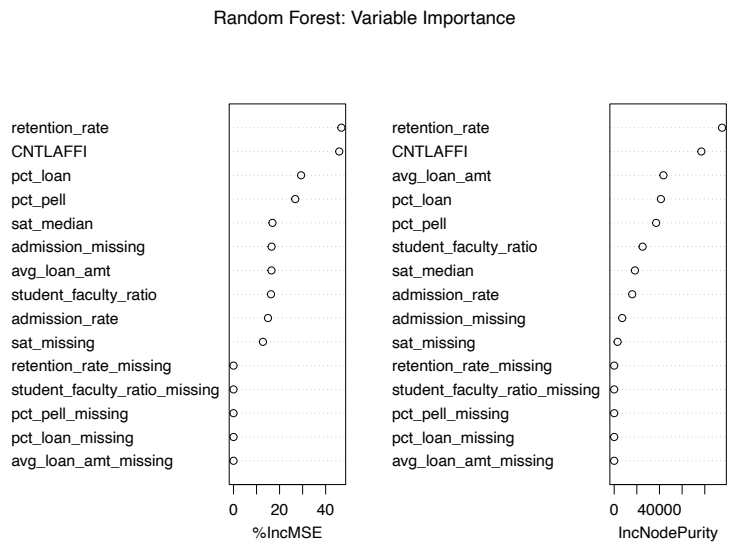

Fig. 9. Random Forest variable importance plot: the relative contribution of predictors to Hispanic completion rates.

Variable importance Fig. 9 revealed three tiers of predictors. The top tier retention rate (%IncMSE = 46.8) and institution type (45.9), accounted for the largest share of model performance, similar to linear regression results. The middle tier included percent loans (29.3), percent Pell Grants (26.8), SAT median (16.9), the admission missing indicator (16.5), average loan amount (16.5), and student-faculty ratio (16.3). While, admission rate (15.0) and the SAT missing indicator (12.8) were the lowest-ranked predictors. The five all zero missing-indicator columns are visible at the bottom of the importance plot and reflect that the loaded modeling dataset had already been imputed.

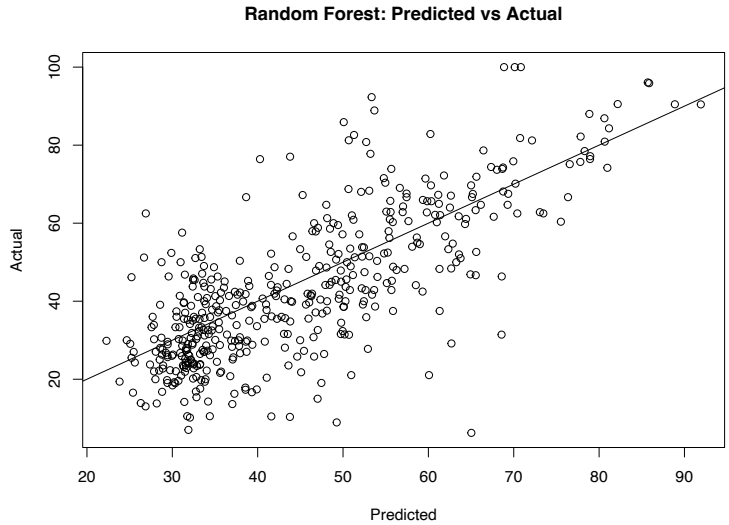

Fig. 10. Test-set predictions vs actual completion rates for the random forest model.

Test-set predictions versus actuals Fig. 9 clusters more tightly around the diagonal than the linear model, with reduced spread at both ends of the predicted range.

To examine the marginal effect of each top predictor while averaging over all others, partial dependence plots (PDPs) were computed for four leading variables Figs. 11–14.

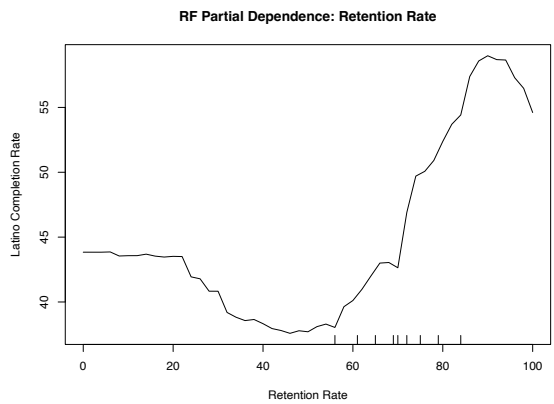

Fig. 11. Random Forest partial dependence on retention rate.

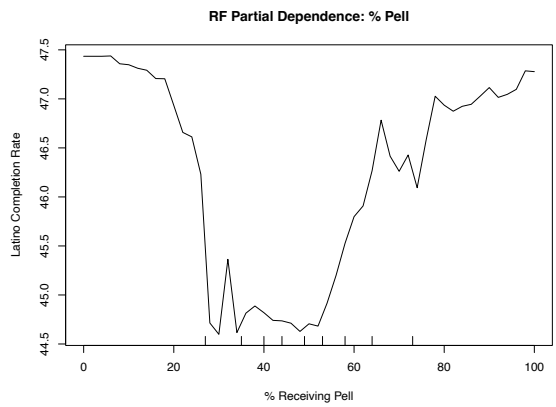

Fig. 13. Random Forest partial dependence on the percentage of students receiving Pell Grants.

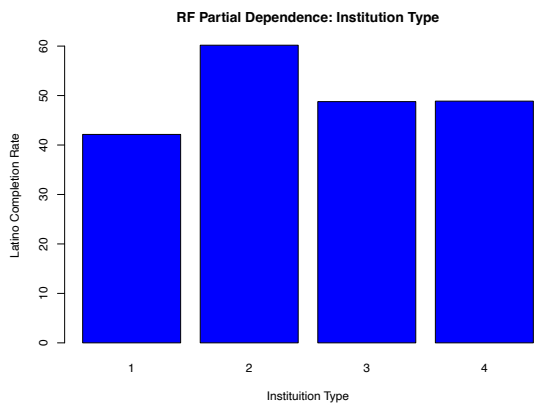

Fig. 12. Random Forest partial dependence on institution type.

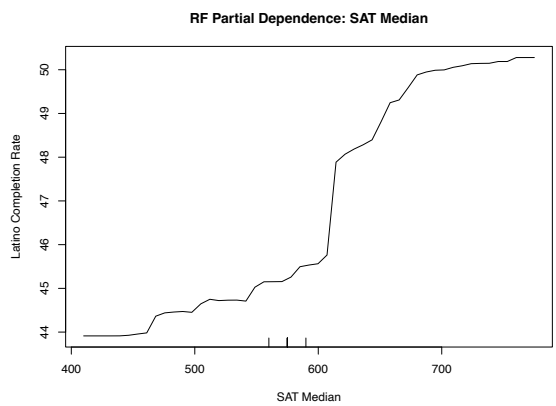

Fig. 14. Random Forest partial dependence on median SAT score.

The retention rate PDP Fig. 11 shows a clear threshold pattern: predicted completion is approximately flat below approx. 50% retention, then rises steeply between approx. 55% and 90%, a nonlinear pattern the linear model cannot capture. The institution type PDP Fig. 12 confirms the for-profit leading against the others previously in linear regression, since private for-profits shows higher predicted completion rate. The SAT median PDP Fig. 14 shows a step increase in predicted completion above approx. 600 on the SAT median.

Interestingly, in the percent Pell Grant PDP Fig. 13, despite a near zero linear correlation ($r = -0.04$) and statistical insignificance in the linear model ($p = 0.73$), percent receiving Pell Grants ranks fourth in

random forest variable importance and demonstrates a clear nonlinear relationship with completion rate in the PDP. Thus, reflecting a nonlinear effect that linear methods cannot detect.

E. Bagging

Bagging was evaluated to determine whether using all predictors at each split, rather than a random subset, could improve performance. In this setting, bagging was implemented as a random forest with $m_{try} = p$ and 500 trees. On the test set, the model achieved MSE = 165.42, RMSE = 12.86, and $R^2 = 0.541$. This was better than both linear regression and elastic net, but slightly worse than random forest.

The result suggests that averaging many trees does improve prediction, but that the random feature restriction used in random forests still adds value. By reducing correlation among trees, random forests generalize slightly better than bagging in this dataset. The bagging importance plot also followed the same overall pattern seen in random forest: retention rate and institution type (CNTLAFFI) were the strongest predictors, followed by financial variables such as percent Pell Grants, percent loans, and average loan amount. SAT and admissions variables contributed less. Overall, bagging supports the same conclusion as random forest, but with slightly lower predictive accuracy.

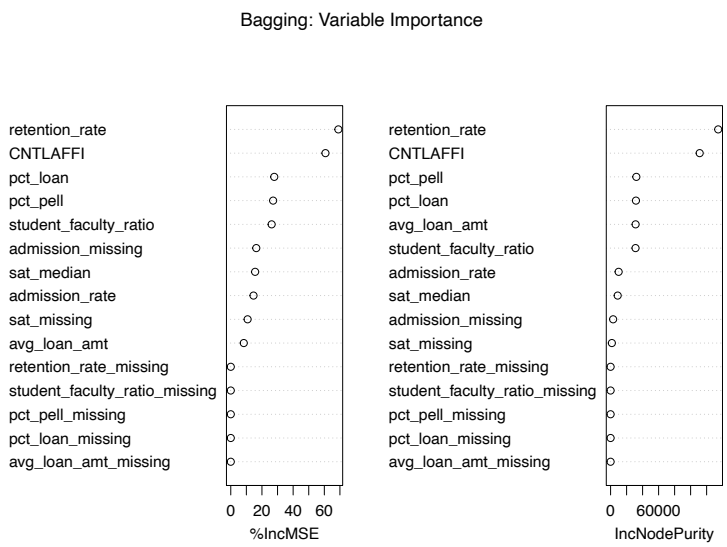

Fig. 15. Bagging variable importance using all 15 predictors at every split (mtry = p).

F. Boosting (GBM)

Gradient boosting was included as a second ensemble approach with a different structure from bagging and random forest. Instead of growing trees independently, boosting builds them sequentially, with each tree fitted to the errors left by the previous stage. The GBM model was trained with 5,000 trees, interaction *depth = 2*, and *shrinkage = 0.01*, with the optimal number of trees selected by 10-fold cross-validation. The procedure selected 1,036 trees.

On the test set, boosting achieved MSE = 167.60, RMSE = 12.95, and $R^2 = 0.535$. This placed it slightly behind both random forest and bagging, while still outperforming the linear methods. In other words, boosting captured useful nonlinear structure, but under the current tuning settings it did not improve enough to overtake random forest. Even so, the variable influence results were consistent with the other

tree models: retention rate and institution type again emerged as the dominant predictors, while financial variables formed a secondary group. This consistency strengthens the conclusion that those variables are the main institutional drivers of Hispanic completion in the current dataset.

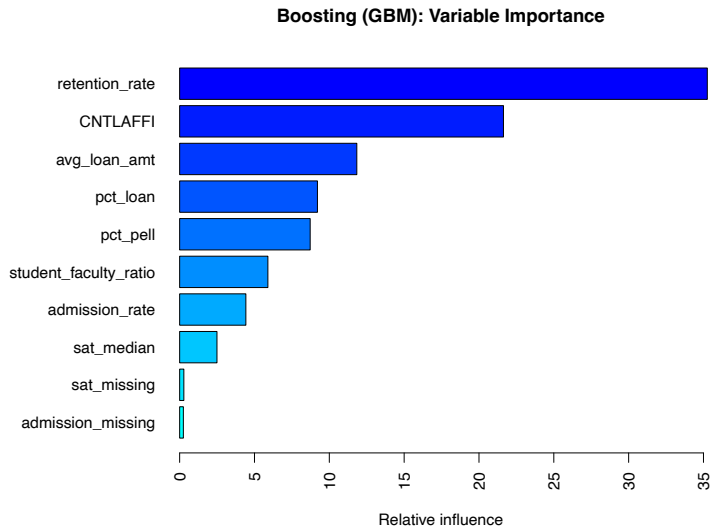

Fig. 16. Gradient Boosting (GBM) relative-influence plot. Predictors are ordered by their proportional contribution to the boosted model.

G. Model Comparison

The test set performance for all five modeling approaches is summarized in Table IV. Random forest achieves the lowest test MSE and RMSE and the highest test R^2, followed closely by bagging and boosting. The two linear methods, linear regression and elastic net, performed lower than the tree-based methods significantly.

TABLE IV. MODEL COMPARISON

Model	Test MSE	Test RMSE	Test R^2
Random Forest	**159.18**	**12.62**	**0.5586**
Bagging	165.42	12.86	0.5413
Boosting	167.60	12.95	0.5353
Elastic Net ($\alpha = 0.1$)	186.15	13.64	0.4838
Linear Regression	186.49	13.66	0.4829

Note: RMSE is in percentage-point units of the Hispanic completion rate. Models are sorted by Test RMSE in ascending order. Final report should bold the highlighted Random Forest row.

To assess whether the RMSE differences in Table IV reflect statistically significant predictive improvements, paired *t*-tests on observation's squared errors were performed comparing random forest with each of the other four models. Results are shown in Table V.

TABLE V. PAIRED T-TEST

Comparison	Mean Δ in SqErr	p-value	Significant
RF vs Linear Regression	−27.31	0.0065	**Yes (**)**
RF vs Elastic Net	−26.97	0.0058	**Yes (**)**
RF vs Bagging	−6.24	0.1468	No
RF vs Boosting	−8.42	0.1065	No

Note: Each test compares the per-observation squared errors of Random Forest against the indicated model on the held-out test set. A negative mean difference indicates that Random Forest had smaller errors on average. Significance codes: ** $p < 0.01$; no marker $p \geq 0.05$.

Random forest's improvement over both linear methods was statistically significant for linear regression and elastic net, confirming that the tree-based methods extract nonlinear and interaction signal more than what the linear models capture. However, random forest's improvement over bagging and boosting was not statistically significant, indicating that the three tree-based ensembles perform comparably on this dataset.

V. CONCLUSIONS AND FUTURE WORK

A. Summary of Findings

In order to predict Hispanic completion rates, five models were evaluated, Linear Regression, Elastic Net, Random Forest, Bagging, and Boosting. Random Forest was the best performing model with MSE = 159.18, RMSE = 12.62, $R^2 = 0.559$ (55.86%). The tree methods, Random Forest, Bagging, and Boosting performed much better than linear approaches, Linear Regression and Elastic Net. Both Elastic Net and Linear Regression performed nearly identical, which reflects how regularization provided no meaningful benefit. Additionally, this pattern indicates that genuine nonlinear interaction exists in the data beyond what linear based methods can capture, but that no ensemble model dominates drastically over the others.

Throughout the models, two predictors were determined as primary drivers: retention rate and institution type. Noted by linear regression coefficients and on random forest, bagging, and boosting variable importance, which are three various tree ensemble methods that differ from one another, which strengthens the idea that these factors are primary drivers.

Notably was the difference between linear and nonlinear evidence on the percent Pell Grant predictor captured. Percent Pell Grant had essentially no linear correlation and was statistically insignificant in the linear regression model, yet was ranked within the top predictors in every tree-based model and showed a clear nonlinear partial dependence shape. Therefore, features that linear regression may not reflect as significant, can be a meaningful nonlinear signal, and ensemble methods provide a way to demonstrate that. While, SAT median score showed the opposite pattern, significant in linear regression model but lower importance in tree models, which can suggest that high academic performance is positively associated with completion but contributes less to predictive accuracy than retention rate or institution type. Financial-aid variables such as percent loan and percent Pell Grant, average loan amount consistently occupied a middle tier of importance, indicating a real but secondary role.

B. Limitations

There were several limitations within our finding. For instance, the analysis is based on a single data set year 2024, so the results cannot be generalized across time and may not be reproducible. Also, the analysis is conducted at the institution level, the model speaks to differences in average outcomes between institutions, not to differences across student groups within institutions. Additionally, considering the importance and observing how these variables interact may be sufficient for descriptive analysis but not to apply as interpretation to make policy changes. Moreover, IPEDS's six-year completion metric counts students who transfer out as noncomplete at their starting institution and excludes part-time enrollment from the cohort definition altogether. Both choices disproportionately affect the Hispanic student population and limit the metric's interpretability. Finally, several other important predictors are not present in the dataset, such as, campus support programming, regional economic context, and student-level financial circumstances, which constrains how much variance any institution level model can explain. Also, the best-performing model's RMSE of 12.62% represents a substantial residual error, that although it is reasonable for analysis, maybe not for prediction.

C. Future Work

There are several future works that follow from this analysis. Such as, extending to multi-year IPEDS data would allow temporal trends to be assessed and test whether these patterns generalize across various years. Also, replicating the methodology on other underrepresented demographics, such as Black and/or Asian students, would help distinguish patterns specific to Hispanic students from broader factors. Also, being able incorporate more predictors not present in IPEDS, to improve predictive accuracy. Lastly, the institutional type for-profit finding could have more analysis to determine bias from genuine institutional effects.

VI. PROJECT REPOSITORY

The code and dataset used for this project is available at: https://github.com/mjgonzo11/R-Statistical-Learning-with-Applications.

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